Linear Statistical Models Worksheet 1

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Due: 19:59, Thursday 20 November 2025

The dataset **satgpa** contains the academic records of 1000 students at an unnamed US college. It contains the following information of each student: first year grade point average at college (variable **fy_gpa**), their verbal SAT percentile (variable **sat_v**), their maths SAT percentile (variable **sat_m**), the total of their SAT percentiles (variable **sat_sum**), their high school grade point average (variable **hs_gpa**) and their sexes (variable **sex**, 1= male, 2= female).

1. Use a multiple regression model to explore the relationship between a person's first year grade point average, their SAT percentile, and their high school grade point average. Comment on the predictive power of the model.

SOLUTION:

We aim to explore whether a student's first year performance at College (University) can be predicted based on their high school SAT and GPA statistics. We begin by fitting the model:

$$fy_gpa_i = \beta_0 + \beta_1 sat_sum_i + \beta_2 hs_gpa_i + \varepsilon_i \quad ; \quad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$$

We can do this in R:

```
md <- lm(fy_gpa ~ sat_sum + hs_gpa, data = satgpa)
```

We then aim to perform an F-test on the predictive power of the model:

$$\mathcal{H}_0 : \beta_1 = \beta_2 = 0 \text{ against } \mathcal{H}_1 : \mathcal{H}_0 \text{ is untrue.}$$

We can do this from the model summary in R:

```
> summary(md)

Call:
lm(formula = fy_gpa ~ sat_sum + hs_gpa)

Residuals:
     Min       1Q   Median       3Q      Max
-2.11889 -0.34126  0.02434  0.40953  1.62368

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.873433   0.149028  -5.861 6.25e-09 ***
sat_sum      0.014402   0.001458   9.878  < 2e-16 ***
hs_gpa       0.579489   0.038456  15.069  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 ''**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5941 on 997 degrees of freedom
Multiple R-squared:  0.3581,    Adjusted R-squared:  0.3568
F-statistic: 278.1 on 2 and 997 DF,  p-value: < 2.2e-16
```

We observe that the p-value of the predictive power test performed is less than $2.2 \cdot 10^{-16}$ - which is the double datatype truncation value for values close to 0 in R. I.e. because the p-value obtained is significantly lower than $\alpha = 5\%$, we can reject the null hypothesis, $\mathcal{H}_0$, and conclude that at least one of the β coefficients for our two regressors, *sat_sum* and *hs_gpa* is non-zero. Therefore, the model has predictive power.

We could also achieve these results using an ANOVA table in R:

```
> anova_larger <- anova(md)
> anova_larger
Analysis of Variance Table

Response: fy_gpa
           Df Sum Sq Mean Sq F value    Pr(>F)
sat_sum     1 116.15 116.150  329.04 < 2.2e-16 ***
hs_gpa      1  80.16  80.156  227.07 < 2.2e-16 ***
Residuals 997 351.94   0.353
---
Signif. codes:  0 '***' 0.001 ''**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> Q = anova_larger$`Sum Sq`[1] + anova_larger$`Sum Sq`[2]
> Q
[1] 196.3065
> M = Q/(anova_larger$`Df`[1] + anova_larger$`Df`[2])
> M
[1] 98.15325
> S2 = (anova_larger$`Sum Sq`[3])/(anova_larger$`Df`[3])
> S2
[1] 0.352996
> F=M/S2
> F
[1] 278.0577
> P=pf(F, anova_larger$`Df`[1] + anova_larger$`Df`[2], anova_larger$`Df`[3], lower.tail =
    FALSE)
> P
[1] 1.082625e-96
```

Notice we observe the same F statistic has been recovered here as was given by the linear model summary, and as we can obtain an exact numerical value from the $F_{2,997}$ distribution in R, we can obtain a more exact p-value of just $1.08 \cdot 10^{-96}$; giving us enormous evidence against $\mathcal{H}_0$ hence supporting the alternative hypothesis of the model having predictive power.

2. Drop the variable **hs_gpa** and fit a model between the **fy_gpa** and **sat_sum**. By constructing a suitable ANOVA table, decide between the model from the first question and the one you just fitted which one you prefer. Include the ANOVA table in your answer.

SOLUTION:

We can construct a second model:

$$fy_gpa_i = \beta_0 + \beta_1 sat_sum_i + \varepsilon_i \quad ; \quad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$$

Which we can then fit in R:

```
md_reduced <- lm(fy_gpa ~ sat_sum, data = satgpa)
```

We then aim to perform the contrast test:

$$\mathcal{H}_0 : \beta_2 = 0 \text{ against } \mathcal{H}_1 : \beta_2 \neq 0 \ \ (\mathcal{H}_0 \text{ is untrue}).$$

We construct an ANOVA table in R, comparing the larger and smaller models:

```
> anova_comparison <- anova(md_reduced,md)
> anova_comparison
Analysis of Variance Table

Model 1: fy_gpa ~ sat_sum
Model 2: fy_gpa ~ sat_sum + hs_gpa
  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
1    998 432.09
2    997 351.94  1    80.156 227.07 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 ''**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

This allows us to form our ANOVA table in the desired format:

```
> df  <- c(1, 997, 998)
> ss  <- c(80.156, 351.94, 432.09)
> ms  <- c(80.156, 351.94/997,"")
> f  <- c((80.156/1)/(351.94/997), "", "")
> anova_table <- cbind(df, ss, ms, f)
> colnames(anova_table) <- c("df", "Sum of squares", "Mean squares", "F")
> rownames(anova_table) <- c("Extra", "Larger", "Smaller")
> as.table(anova_table)
        df  Sum of squares Mean squares      F
Extra   1   80.156         80.156           227.071466727283
Larger  997 351.94         0.352998996990973
Smaller 998 432.09
> pf(227.0715, 1, 997, lower.tail = FALSE)
[1] 2.209382e-46
```

The obtained p-value of less than $2.2 \cdot 10^{-16}$ (determined to be exactly $2.2 \cdot 10^{-46}$) is significantly less that 5% which therefore means that we can confidently reject the null hypothesis, $\mathcal{H}_0$. Hence we are able to conclude that a student's high school GPA is a statistically significant predictor of a their GPA in their first year of college, given their high school SAT scores. Therefore, we prefer the first, larger model which includes high school GPA.

3. The total SAT percentile corresponds to the sum of verbal and maths SAT percentiles. Now fit a model with **fy_gpa** as the response variable and **sat_v**, **sat_m** and **hs_gpa** as regressors. Compared with the model fitted in the first question, which one do you prefer? Explain your reasons.

SOLUTION:

We can construct a third model:

$$fy_gpa_i = \beta_0 + \beta_1 sat_v_i + \beta_2 sat_m_i + \beta_3 hs_gpa_i + \varepsilon_i \quad ; \quad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$$

Which we can then fit in R:

```
md2 <- lm(fy_gpa ~ sat_m + sat_v + hs_gpa, data = satgpa)
```

We then aim to test whether:

$$\mathcal{H}_0 : \beta_1 = \beta_2 \text{ (model 1) against } \mathcal{H}_1 : \beta_1 \neq \beta_2 \text{ } (\mathcal{H}_0 \text{ is untrue}).$$

Or equivalently:

$$\mathcal{H}_0 : \beta_1 - \beta_2 = 0 \text{ (model 1) against } \mathcal{H}_1 : \beta_1 - \beta_2 \neq 0 \text{ } (\mathcal{H}_0 \text{ is untrue}).$$

Since under $\mathcal{H}_0$, we have:

$$\begin{aligned} fy_gpa_i &= \beta_0 + \beta_1 sat_v_i + \beta_2 sat_m_i + \beta_3 hs_gpa_i + \varepsilon_i \\ &= \beta_0 + \beta_1 sat_v_i + \beta_1 sat_m_i + \beta_3 hs_gpa_i + \varepsilon_i \\ &= \beta_0 + \beta_1 \left(sat_v_i + sat_m_i\right) + \beta_3 hs_gpa_i + \varepsilon_i \\ &= \beta_0 + \beta_1 \left(sat_sum_i\right) + \beta_3 hs_gpa_i + \varepsilon_i \\ &= \beta_0 + \beta_{\text{sum}} \left(sat_sum_i\right) + \beta_3 hs_gpa_i + \varepsilon_i \end{aligned}$$

Note, we can reparameterise the model as follows:

Since $sat_sum_i := sat_v_i + sat_m_i$, we can define $sat_diff_i := sat_v_i - sat_m_i$.

Then it follows that:

$$sat_v_i = \tfrac{1}{2}\left(sat_sum_i + sat_diff_i\right) \text{ and } sat_m_i = \tfrac{1}{2}\left(sat_sum_i - sat_diff_i\right).$$

Substituting this into our model yields:

$$fy_gpa_i = \beta_0 + \tfrac{\beta_1+\beta_2}{2}\, sat_sum_i + \tfrac{\beta_1-\beta_2}{2}\, sat_diff_i + \beta_3\, hs_gpa_i + \varepsilon_i.$$

$$= \beta_0 + \beta_{\text{sum}}\, sat_sum_i + \beta_{\text{diff}}\, sat_diff_i + \beta_3\, hs_gpa_i + \varepsilon_i.$$

Comparing this to our first model:

$$fy_gpa_i = \beta_0 + \beta_1 sat_sum_i + \beta_2 hs_gpa_i + \varepsilon_i$$

We see that our hypothesis test now becomes:

$$\mathcal{H}_0 : \beta_{\text{diff}} = 0 \text{ (model 1) against } \mathcal{H}_1 : \beta_{\text{diff}} \neq 0 \ \ (\mathcal{H}_0 \text{ is untrue}).$$

Since under our reparameterisation, we have that the test $\beta_{\text{diff}} = 0$ is equivalent to testing $\frac{1}{2}(\beta_1 - \beta_2) = 0$. Which is then equivalent to testing $\beta_1 - \beta_2 = 0$ - which is precisely the original hypothesis test under $\mathcal{H}_0$.

Under the alternative hypothesis, $\mathcal{H}_1$, we are stating that the term β_{diff}, which represents the effect of a student's difference in verbal and maths SAT scores on their first year college GPA, is statistically significant (non-zero).

This now becomes a simple model comparison test in R:

```
> md <- lm(fy_gpa ~ sat_sum + hs_gpa, data = satgpa)
> md2 <- lm(fy_gpa ~ sat_m + sat_v + hs_gpa, data = satgpa)
> anova(md,md2)
Analysis of Variance Table

Model 1: fy_gpa ~ sat_sum + hs_gpa
Model 2: fy_gpa ~ sat_m + sat_v + hs_gpa
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1    997 351.94
2    996 351.63  1   0.30562 0.8657 0.3524
> satgpa$sat_diff <- satgpa$sat_v - satgpa$sat_m
> md_diff <- lm(fy_gpa ~ sat_sum + sat_diff + hs_gpa, data = satgpa)
> anova(md,md_diff)
Analysis of Variance Table

Model 1: fy_gpa ~ sat_sum + hs_gpa
Model 2: fy_gpa ~ sat_sum + sat_diff + hs_gpa
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1    997 351.94
2    996 351.63  1   0.30562 0.8657 0.3524
```

For the sake of completeness, both model parameterisations are included, although we observe they produce the same results, as we expected.

We observe now that this model comparison test produced a p-value of 0.3524 which is sufficiently greater than 5% to provide a lack of evidence that the null hypothesis, $\mathcal{H}_0$ should be rejected. Therefore, there is insufficient evidence to conclude that $\beta_{\text{diff}} \neq 0 \iff \beta_1 \neq \beta_2$.

In conclusion, we will prefer to use our original model:

$$fy_gpa_i = \beta_0 + \beta_1 sat_sum_i + \beta_2 hs_gpa_i + \varepsilon_i$$

Further justifications to prefer the simpler first model include:

- A reduced error variance accompanied by an increase in the adjusted coefficient of determination, R_a^2 - although both changes are negligible. We observe this on the following R output of the model summaries, whereby $(\hat{\sigma}^2, R_a^2)_{\text{model 1}} = (0.5941^2, 0.3568)$ and $(\hat{\sigma}^2, R_a^2)_{\text{model 2}} = (0.5942^2, 0.3567)$:

```
> md <- lm(fy_gpa ~ sat_sum + hs_gpa, data = satgpa)
> summary(md)

Call:
lm(formula = fy_gpa ~ sat_sum + hs_gpa, data = satgpa)

Residuals:
     Min       1Q   Median       3Q      Max
```

```
-2.11889 -0.34126  0.02434  0.40953  1.62368

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.873433   0.149028  -5.861 6.25e-09 ***
sat_sum      0.014402   0.001458   9.878  < 2e-16 ***
hs_gpa       0.579489   0.038456  15.069  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 ''**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5941 on 997 degrees of freedom
Multiple R-squared:  0.3581,    Adjusted R-squared:  0.3568
F-statistic: 278.1 on 2 and 997 DF,  p-value: < 2.2e-16

> md2 <- lm(fy_gpa ~ sat_m + sat_v + hs_gpa, data = satgpa)
> summary(md2)

Call:
lm(formula = fy_gpa ~ sat_m + sat_v + hs_gpa, data = satgpa)

Residuals:
     Min       1Q   Median       3Q      Max
-2.10154 -0.35893  0.02541  0.41321  1.61394

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.866927   0.149202  -5.810 8.38e-09 ***
sat_m        0.012398   0.002601   4.766 2.16e-06 ***
sat_v        0.016459   0.002648   6.215 7.55e-10 ***
hs_gpa       0.580068   0.038463  15.081  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 ''**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5942 on 996 degrees of freedom
Multiple R-squared:  0.3586,    Adjusted R-squared:  0.3567
F-statistic: 185.6 on 3 and 996 DF,  p-value: < 2.2e-16
```

Therefore, the essentially unchanged error variance and adjusted R^2 indicates negligible improvement from the more complex model.

- Ockham's razor suggests that we should prefer model 1 given that it is built on fewer regressors.
- Since model 1 does not differentiate maths and verbal SAT scores, it avoids any possible multicollinearity between the two score types.

4. By adding **sex** as a qualitative variable into the model fitted in the first question, investigate whether sex is a factor in influencing one's first year GPA when SAT and high school GPA have been taken into account.

SOLUTION:

We can construct a fourth model (for now assuming no interaction):

$$fy_gpa_{ij} = \mu + \underbrace{\alpha_i}_{\substack{\text{Effect of} \\ i^{\text{th}} \text{ sex}}} + \beta_1 \underbrace{sat_sum_{ij}}_{\substack{\text{SAT score of } j^{\text{th}} \\ \text{person in } i^{\text{th}} \text{ sex}}} + \beta_2 \underbrace{hs_gpa_{ij}}_{\substack{\text{HS GPA of } j^{\text{th}} \\ \text{person in } i^{\text{th}} \text{ sex}}} + \varepsilon_{ij} \;\; ; \;\; \varepsilon_{ij} \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad i = \begin{cases} 1 & : \text{Male} \\ 2 & : \text{Female} \end{cases}$$

Under this model we would want to test:

$$\mathcal{H}_0 : \alpha_1 = \alpha_2 \text{ against } \mathcal{H}_1 : \alpha_1 \neq \alpha_2 \;\; (\mathcal{H}_0 \text{ is untrue}).$$

Which we can then fit in R:

```
> md3 <- lm(fy_gpa ~ sat_sum + hs_gpa + factor(sex), data = satgpa)
> summary(md3)

Call:
lm(formula = fy_gpa ~ sat_sum + hs_gpa + factor(sex), data = satgpa)

Residuals:
```

```
     Min       1Q   Median       3Q      Max
-2.06456 -0.34544  0.03185  0.41002  1.70640

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  -0.978893   0.150887  -6.488 1.37e-10 ***
sat_sum       0.015831   0.001501  10.550  < 2e-16 ***
hs_gpa        0.544622   0.039388  13.827  < 2e-16 ***
factor(sex)2  0.143059   0.039089   3.660 0.000266 ***
---
Signif. codes:  0 '***' 0.001 ''**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5905 on 996 degrees of freedom
Multiple R-squared:  0.3666,    Adjusted R-squared:  0.3647
F-statistic: 192.1 on 3 and 996 DF,  p-value: < 2.2e-16
```

Under our additive model, it seems that the qualitative factor *sex* is statistically significant, since the p-value associated with the coefficient of `factor(sex)2` (0.000266) is sufficiently smaller than 5%. Therefore, there is sufficient evidence to reject the null hypothesis $\mathcal{H}_0 : \alpha_1 = \alpha_2$, and hence, given *sat_sum* and *hs_gpa*, the mean first year GPA differs between males and females. However, we then need to test for interactions using the following model:

$$fy_gpa_{ij} = \mu + \alpha_i + \beta_1\, sat_sum_{ij} + \beta_2\, hs_gpa_{ij} + \mathbb{1}_{\{i=2\}}\left(\delta_1\, sat_sum_{ij} + \delta_2\, hs_gpa_{ij}\right) + \varepsilon_{ij}$$

$$\implies \begin{cases} \text{Male } (i=1): & fy_gpa_{1j} = \mu + \alpha_1 + \beta_1\, sat_sum_{1j} + \beta_2\, hs_gpa_{1j} + \varepsilon_{1j}, \\ \text{Female } (i=2): & fy_gpa_{2j} = \mu + \alpha_2 + (\beta_1+\delta_1)\, sat_sum_{2j} + (\beta_2+\delta_2)\, hs_gpa_{2j} + \varepsilon_{2j}. \end{cases}$$

To check if either of the interactions terms are present, we need to perform the following ANCOVA test:

$$\mathcal{H}_0 :\ \delta_1 = \delta_2 = 0 \quad \text{against} \quad \mathcal{H}_1 :\ \text{at least one of } \delta_1, \delta_2 \neq 0,$$

Which can be done using an F-test against the additive (no interaction) model we defined. I.e. we perform an anova test in R as follows:

```
> md3 <- lm(fy_gpa ~ sat_sum + hs_gpa + factor(sex), data = satgpa)
> md4 <- lm(fy_gpa ~ (sat_sum + hs_gpa) * factor(sex), data = satgpa)
> anova(md3,md4)
Analysis of Variance Table

Model 1: fy_gpa ~ sat_sum + hs_gpa + factor(sex)
Model 2: fy_gpa ~ (sat_sum + hs_gpa) * factor(sex)
  Res.Df    RSS Df Sum of Sq      F   Pr(>F)
1    996 347.27
2    994 343.92  2    3.3429 4.8308 0.008168 **
---
Signif. codes:  0 '***' 0.001 ''**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The F-test comparing the additive and interaction models gives significant evidence that interaction is present due to a p-value of 0.00817. Hence we have sufficient evidence to reject the null hypothesis, $\mathcal{H}_0$, and conclude that there is interaction present in this model.

Next, we want to determine what kind of interaction is present, since the initial ANCOVA test only told us that at least one of $\delta_1, \delta_2 \neq 0$. We can observe the model's summary in R:

```
> summary(md4)

Call:
lm(formula = fy_gpa ~ (sat_sum + hs_gpa) * factor(sex), data = satgpa)

Residuals:
     Min       1Q   Median       3Q      Max
-2.06933 -0.35171  0.03089  0.39466  1.66355

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
```

```
(Intercept)          -0.787018   0.209613  -3.755 0.000184 ***
sat_sum               0.017166   0.002067   8.306  3.2e-16 ***
hs_gpa                0.438058   0.052234   8.387  < 2e-16 ***
factor(sex)2         -0.297541   0.296326  -1.004 0.315574
sat_sum:factor(sex)2 -0.003339   0.002999  -1.113 0.265822
hs_gpa:factor(sex)2   0.244113   0.079145   3.084 0.002096 **
---
Signif. codes:  0 '***' 0.001 ''**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5882 on 994 degrees of freedom
Multiple R-squared:  0.3727,    Adjusted R-squared:  0.3695
F-statistic: 118.1 on 5 and 994 DF,  p-value: < 2.2e-16
```

This tells us that there is interaction present between *sex* and *hs_gpa*. However, interaction is not present at any reasonable confidence level between *sex* and *sat_sum*.

To confirm this, we might want to perform some ANOVA comparisons. Firstly by defining in R two new models which respectively consider the cases $\delta_1 = 0$ and $\delta_2 = 0$:

```
md_hsgpa  <- lm(fy_gpa ~ sat_sum + hs_gpa + factor(sex) + hs_gpa:factor(sex), data = satgpa)
md_sat <- lm(fy_gpa ~ sat_sum + hs_gpa + factor(sex) + sat_sum:factor(sex), data = satgpa)
```

We can now perform the following comparisons:

1. We firstly compare

   ```
   md3
   ```

   with

   ```
   md_hsgpa
   ```

   which will confirm whether the addition of an interaction term between high school gpa and sex to the additive model is supported.

   ```
   > anova(md3, md_hsgpa)
   Analysis of Variance Table

   Model 1: fy_gpa ~ sat_sum + hs_gpa + factor(sex)
   Model 2: fy_gpa ~ sat_sum + hs_gpa + factor(sex) + hs_gpa:factor(sex)
     Res.Df    RSS Df Sum of Sq    F   Pr(>F)
   1    996 347.27
   2    995 344.35  1     2.914 8.42 0.003793 **
   ---
   Signif. codes:  0 '***' 0.001 ''**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
   ```

   The p-value of 0.003793 gives us strong confidence that there is indeed interaction between high school gpa and sex.

2. We secondly compare

   ```
   md3
   ```

   with

   ```
   md_sat
   ```

   which will confirm whether the addition of an interaction term between high school SAT scores and sex to the additive model is supported.

   ```
   > anova(md3, md_sat)
   Analysis of Variance Table

   Model 1: fy_gpa ~ sat_sum + hs_gpa + factor(sex)
   Model 2: fy_gpa ~ sat_sum + hs_gpa + factor(sex) + sat_sum:factor(sex)
     Res.Df    RSS Df Sum of Sq      F Pr(>F)
   1    996 347.27
   2    995 347.22  1  0.051328 0.1471 0.7014

   ```

As we expected to see here, the p-value of 0.7014 tells us that there is no statistical evidence of an interaction between high school SAT scores and sex.

3. We finally compare

```
md_hsgpa
```

with

```
md4
```

which will confirm whether the removal of the interaction term between high school SAT scores and sex from the full interaction model, leaving only an interaction term between high school gpa and sex is supported.

```
> anova(md_hsgpa, md4)
Analysis of Variance Table

Model 1: fy_gpa ~ sat_sum + hs_gpa + factor(sex) + hs_gpa:factor(sex)
Model 2: fy_gpa ~ (sat_sum + hs_gpa) * factor(sex)
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1    995 344.35
2    994 343.92  1   0.42889 1.2396 0.2658

```

As we expected to see here, the p-value of 0.2658 tells us that there is no statistical evidence of an interaction between high school SAT scores and sex.

Therefore we conclude that our preferred model has an interaction between high school GPA and sex, namely $\delta_1 = 0$, $\delta_2 \neq 0$:

$$\implies fy_gpa_{ij} = \mu + \alpha_i + \beta_1\, sat_sum_{ij} + \beta_2\, hs_gpa_{ij} + \mathbb{1}_{\{i=2\}}\left(\delta_2\, hs_gpa_{ij}\right) + \varepsilon_{ij}$$

$$\implies \begin{cases} \text{Male } (i=1): & fy_gpa_{1j} = \mu + \alpha_1 + \beta_1\, sat_sum_{1j} + \beta_2\, hs_gpa_{1j} + \varepsilon_{1j}, \\ \text{Female } (i=2): & fy_gpa_{2j} = \mu + \alpha_2 + \beta_1\, sat_sum_{2j} + (\beta_2 + \delta_2)\, hs_gpa_{2j} + \varepsilon_{2j}, \end{cases}$$

Whereby, our δ_2 term represents the delta in slope of high school gpa for females relative to males given SAT scores. We can estimate this term from the model summary in R:

```
> summary(md_hsgpa)

Call:
lm(formula = fy_gpa ~ sat_sum + hs_gpa + factor(sex) + hs_gpa:factor(sex),
    data = satgpa)

Residuals:
    Min      1Q  Median      3Q     Max
-2.0674 -0.3520  0.0317  0.3961  1.6245

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)        -0.673991   0.183411  -3.675 0.000251 ***
sat_sum             0.015580   0.001498  10.403  < 2e-16 ***
hs_gpa              0.455530   0.049826   9.142  < 2e-16 ***
factor(sex)2       -0.508466   0.227883  -2.231 0.025886 *
hs_gpa:factor(sex)2  0.202599   0.069820   2.902 0.003793 **
---
Signif. codes:  0 '***' 0.001 ''**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5883 on 995 degrees of freedom
Multiple R-squared:  0.3719,    Adjusted R-squared:  0.3694
F-statistic: 147.3 on 4 and 995 DF,  p-value: < 2.2e-16

```

This gives us as estimate of $\delta_2 = 0.202599$. By observing the summary of this model, we can also note that all terms remain significant with p-values less than 5%.

From this summary, we can construct the fitted values of fy_gpa_{2j} and fy_gpa_{1j}:

$$\begin{cases} \text{Male } (i=1): & \widehat{fy_gpa}_{1j} = -0.674 \ + \ 0.016\, sat_sum_{1j} \ + \ 0.456\, hs_gpa_{1j}, \\ \text{Female } (i=2): & \widehat{fy_gpa}_{2j} = \big(-0.674 - 0.508\big) \ + \ 0.016\, sat_sum_{2j} \ + \ \big(0.456 + 0.203\big)\, hs_gpa_{2j} \\ & \qquad = -1.182 \ + \ 0.016\, sat_sum_{2j} \ + \ 0.658\, hs_gpa_{2j}. \end{cases}$$

The implications of this result are that taking the difference in fitted values of first year GPA between female, and male students yields:

$$\begin{aligned} \Delta = \widehat{fy_gpa}_{2j} - \widehat{fy_gpa}_{1j} &= (\alpha_2 - \alpha_1) + \delta_2 hs_gpa \\ \implies \widehat{\Delta} = \widehat{(\alpha_2 - \alpha_1)} + \widehat{\delta}_2 hs_gpa & \\ = -0.508466 + 0.202599 \cdot hs_gpa & \end{aligned}$$

Which is representative of the interaction found between sex and high school GPA. We can observe this relationship through plots obtained with the following R code:

```
> plot(fy_gpa ~ hs_gpa, data = satgpa, xlab = "High school GPA", ylab = "First year GPA")
>
> male_gpa <- data.frame(hs_gpa = range(satgpa$hs_gpa), sat_sum = mean(satgpa$sat_sum), sex
    = factor(1, levels = c(1, 2)))
> female_gpa <- data.frame(hs_gpa = range(satgpa$hs_gpa),sat_sum = mean(satgpa$sat_sum), sex
        = factor(2, levels = c(1, 2)))
>
> lines(range(satgpa$hs_gpa), predict(md_hsgpa, newdata = male_gpa))
> lines(range(satgpa$hs_gpa), predict(md_hsgpa, newdata = female_gpa), lty = 2)
> legend("topleft", c("Male", "Female"), lty = c(1, 2))
>
> plot(fy_gpa ~ sat_sum, data = satgpa, xlab = "Total SAT score", ylab = "First year GPA")
>
> male_sat <- data.frame(sat_sum = range(satgpa$sat_sum),hs_gpa = mean(satgpa$hs_gpa),sex =
    factor(1, levels = c(1, 2)))
> female_sat <- data.frame(sat_sum = range(satgpa$sat_sum),hs_gpa = mean(satgpa$hs_gpa),sex
    = factor(2, levels = c(1, 2)))
>
> lines(range(satgpa$sat_sum), predict(md_hsgpa, newdata = male_sat))
> lines(range(satgpa$sat_sum), predict(md_hsgpa, newdata = female_sat), lty = 2)
> legend("topleft", c("Male", "Female"), lty = c(1, 2))
```

We plot the predicted first year gpas for males and females separately across two plots which both fix the second respective regressor to its mean from satgpa:

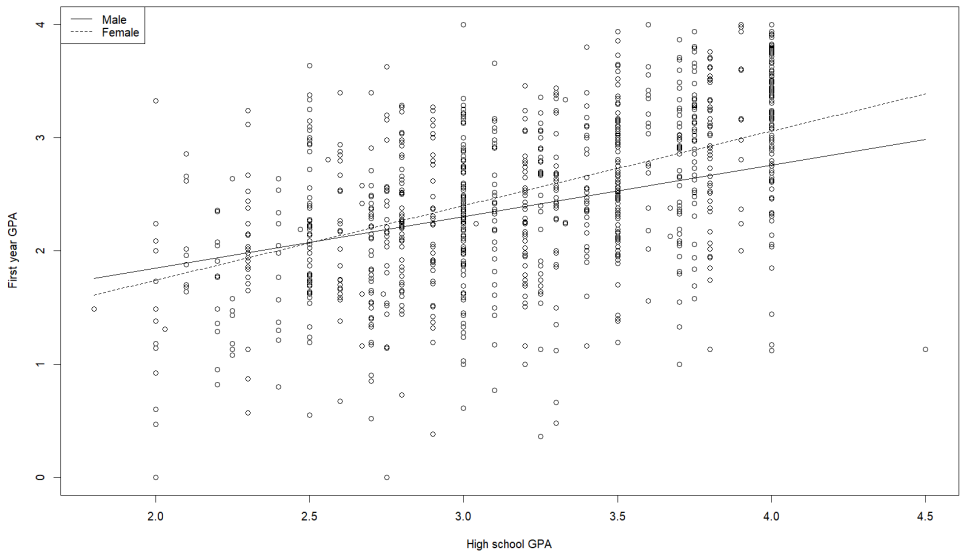


Figure 1: High School GPA vs Fitted First Year GPA linear relationship comparison between males and females given fixed SAT Value of satgpa mean.

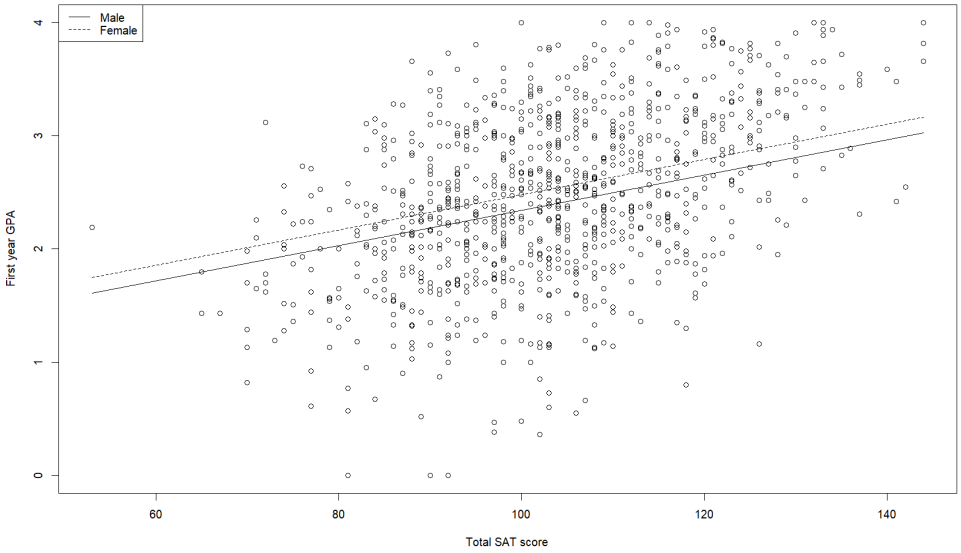


Figure 2: SAT Score vs First Year GPA linear relationship comparison between males and females given fixed GPA Value of satgpa mean.

Given we have shown $\widehat{\Delta} = -0.508466 + 0.202599 \cdot hs_gpa$, we can observe in Figure 1 that since $\widehat{\Delta} = 0 \iff = -0.508466 + 0.202599 \cdot hs_gpa = 0 \implies hs_gpa \approx 2.51$, that is where our straight lines of male and female first year gpas intersect.

Since $\widehat{\Delta} \perp\!\!\!\perp sat_sum$, given Figure 2 is based on a fixed value of hs_gpa equal to its mean in satgpa, $\widehat{\Delta}$ is constant which therefore gives us parallel lines in Figure 2.

From the findings we have made, we can conclude that sex is indeed a factor in influencing students' first year GPA once SAT and high school GPA have been taken into account, and furthermore there is interaction present between sex and high school GPA, which produces non-parallel fitted regression lines for males and females in Figure 1. By contrast, because there is no interaction present between sex and SAT scores, the fitted regression lines in the plot of SAT scores vs first year GPA for males and females are parallel.

5. Are the model assumptions justifiable given the dataset?

SOLUTION:

We will perform a model validation of the model we have determined to provide the best fit to the data:

```
md_hsgpa  <- lm(fy_gpa ~ sat_sum + hs_gpa + factor(sex) + hs_gpa:factor(sex), data = satgpa)
```

To do this, we will need to consider whether the standardised residuals are consistent with $\mathcal{N}(0,1)$ and are sufficiently random implying independence.

1. We firstly notice from the following R code that approximately 95% of the standardised residuals lie in the interval $(-2,2)$ which is roughly what we would expect under $\mathcal{N}(0,1)$.

```
> stdres <- rstandard(md_hsgpa)
> proportion_abs_stdres_less_2 <- mean(abs(stdres) <= 2)
> proportion_abs_stdres_less_2
[1] 0.952
```

2. We can now assess whether the residuals plotted against their index number generated by the following R code display any form of pattern (e.g. funnelling) which could indicate a lack of independence.

```
plot(stdres,xlab = "Index",ylab = "Standardised residuals", main = "satgpa data
    indices vs standardised residuals")
abline(h = 0)
```

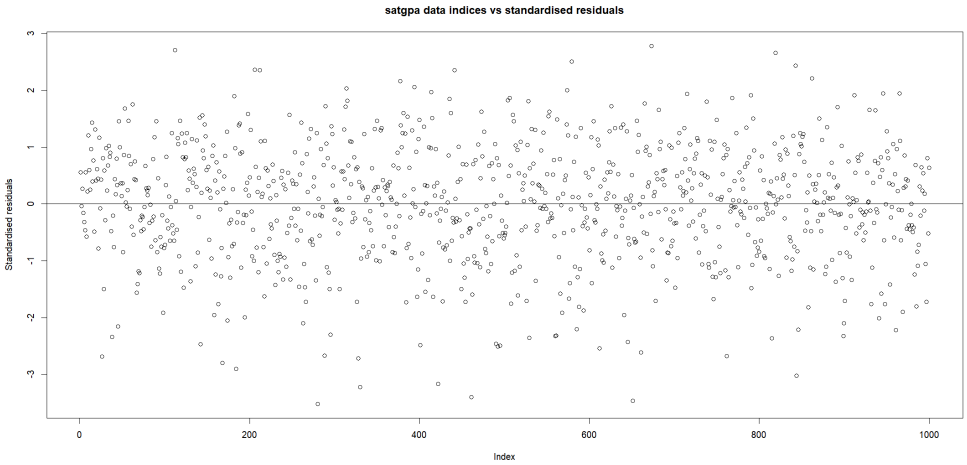


Figure 3: satgpa data indices vs standardised residuals

Figure 3 demonstrates that the standardised residuals plotted against their indices are randomly scattered about zero with no apparent patterns, hence the assumption of error independence seems to be upheld.

3. We can now assess whether the residuals demonstrate constant variance by plotting them against each regressor and the fitted values. Each plot is accompanied by its corresponding block of R code.

   Standardised residuals vs *sat_sum*:

```
plot(satgpa$sat_sum, stdres,xlab = "satgpa$sat_sum",ylab = "Standardised residuals",
    main = "SAT Totals vs Standardised Residuals")
abline(h = 0)
```

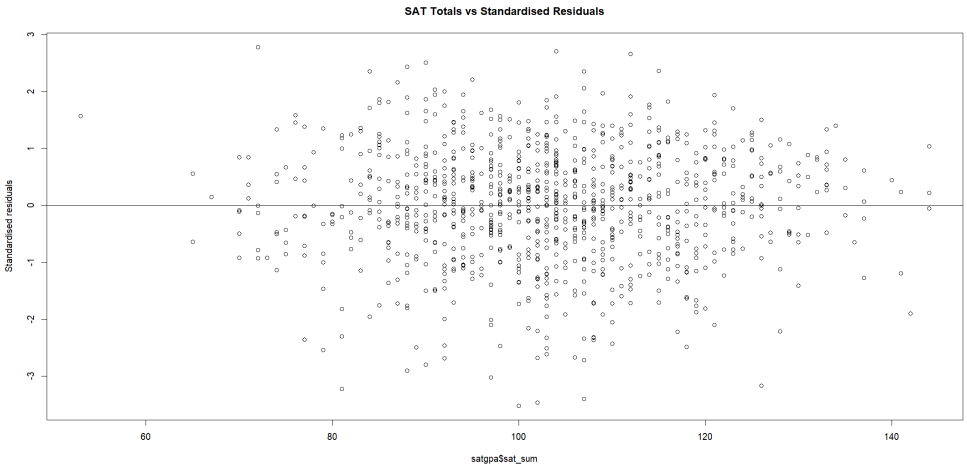


Figure 4: SAT Totals vs Standardised Residuals

Standardised residuals vs *hs_gpa*:

```
plot(satgpa$hs_gpa, stdres,xlab = "satgpa$hs_gpa",ylab = "Standardised residuals",main
     = "High School GPA vs Standardised Residuals")
abline(h = 0)
```

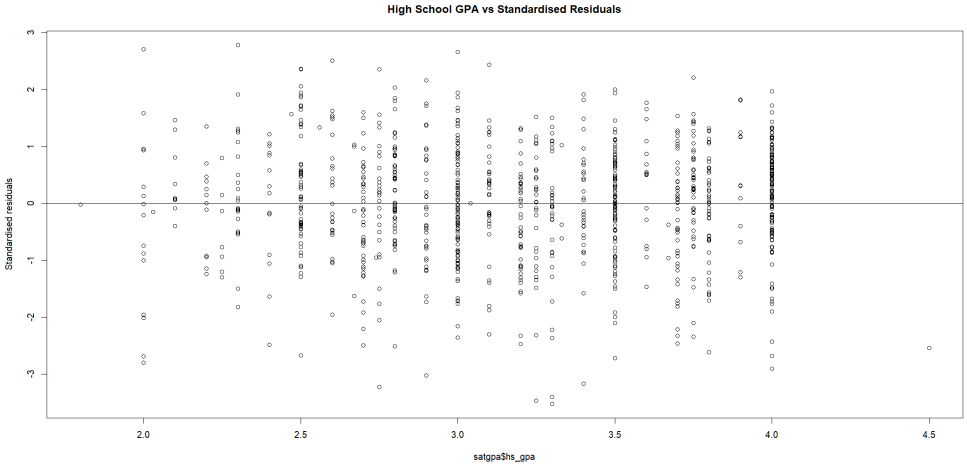


Figure 5: High School GPA vs Standardised Residuals.

Standardised residuals vs fitted values:

```
plot(md_hsgpa$fitted.values, stdres, xlab = "md_hsgpa$fitted.values", ylab ="
    Standardised residuals",main = "Model fitted values vs Standardised Residuals")
abline(h = 0)
```

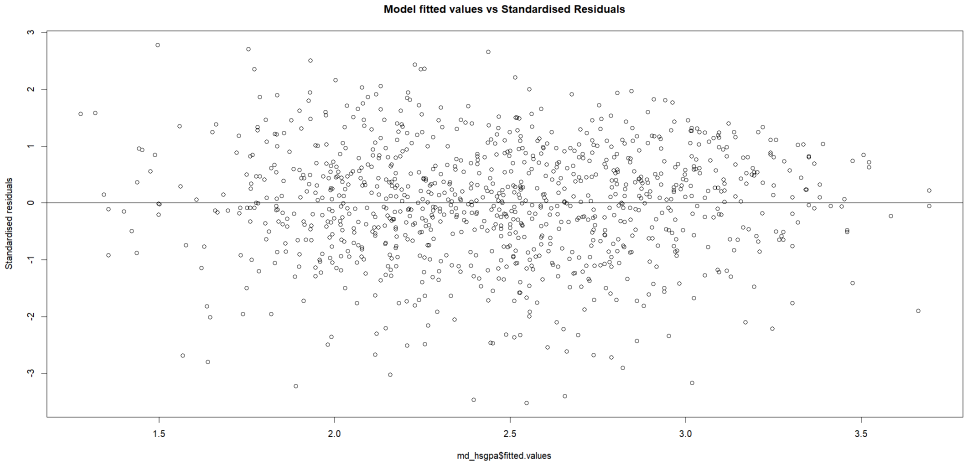


Figure 6: Model fitted values vs Standardised Residuals

Figures 4,5,6 demonstrate even spread of standardised residuals about zero within a fairly uniform band since as we showed, most values fall within the interval $(-2, 2)$. Since there does not seem to be any pattern to the distribution or change in vertical spread of the standardised residuals in any of these plots, we can conclude that the assumption of constant error variance is upheld.

4. We now want to confirm the normality assumption by producing a Normal Probability (QQ) plot and a histogram of the standardised residuals in R:

```
qqnorm(stdres, main = "Normal QQ plot")
qqline(stdres)

hist(stdres,30,xlab = "Standardised residuals",main = "Histogram of standardised
    residuals")
```

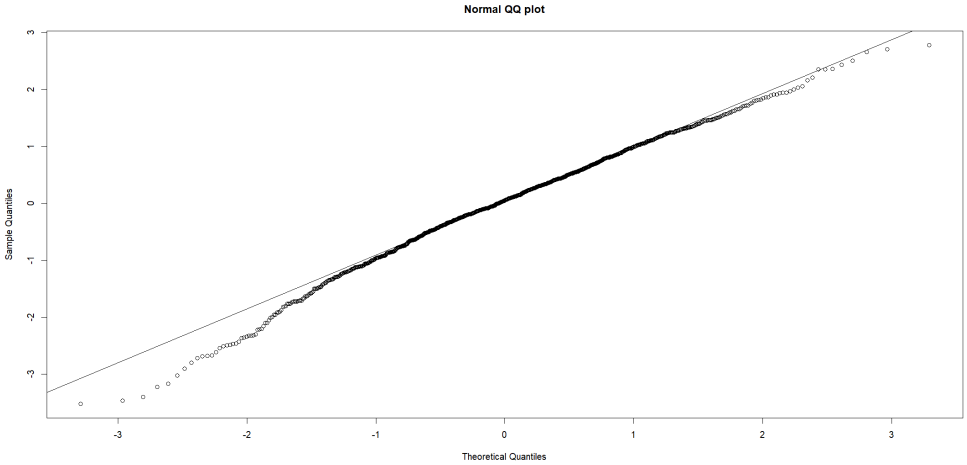


Figure 7: Normal QQ plot.

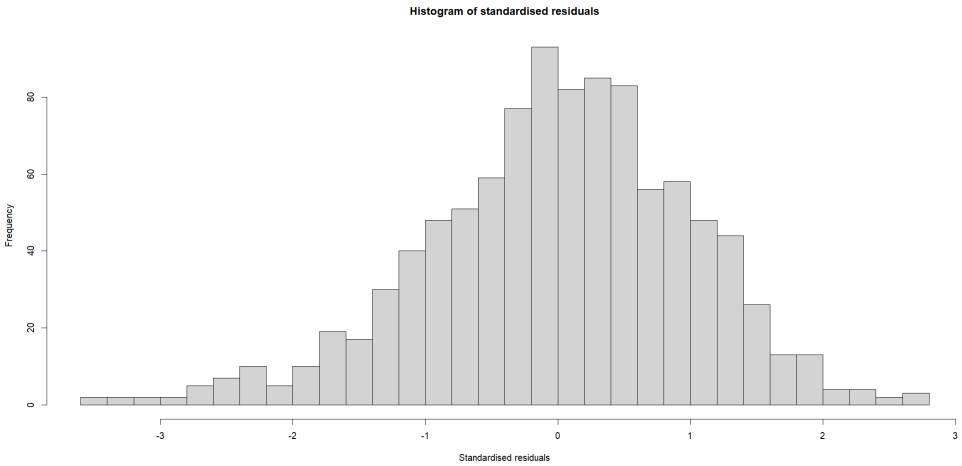


Figure 8: Histogram of standardised residuals.

We observe in Figure 7 that the points on the QQ plot all seem to lie on, or close to the straight line, with only minor deviations in the tails near -1.5 and 2. Then in Figure 8 we observe that the histogram is both roughly symmetric and bell-shaped. Since we have confirmed that approximately 95% of the standardised residuals lie in the interval $(-2, 2)$, this provides strong support of the assumption that they follow $\mathcal{N}(0, 1)$.

5. Finally, we want to consider whether the error variance is constant between the two levels of the sex factor using a boxplot of standardised residuals by sex:

```
boxplot(stdres ~ factor(sex), data = satgpa, xlab = "Sex : 1 = Male, 2 = Female", ylab
    = "Standardised residuals", main = "Boxplot of Sex vs Standardised residuals")
```

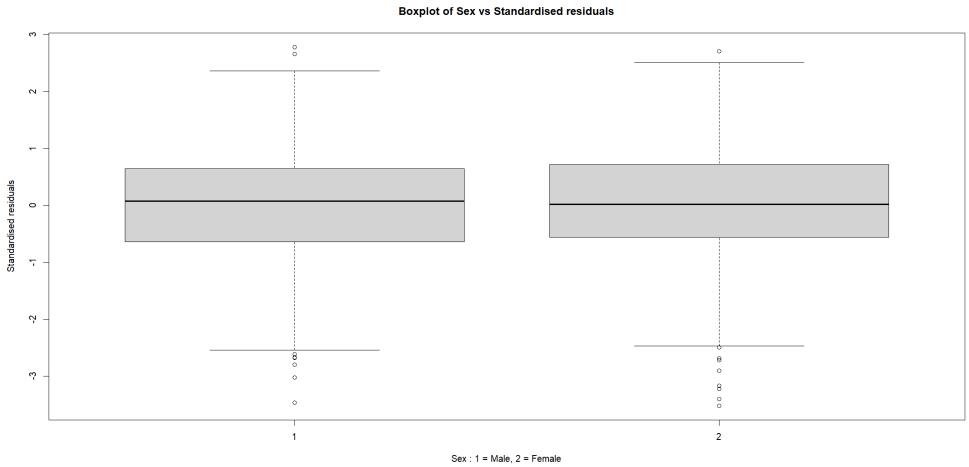


Figure 9: Boxplot of Sex vs Standardised residuals

We can see in Figure 9 that both levels of the factor sex have medians very close to zero, and similar interquartile ranges and variance. This supports the assumption of constant variance in standardised residuals across both levels of the qualitative factor sex in our model.

In conclusion, the standardised residuals appear to be consistent with a Normal distribution with mean zero and constant variance, i.e. $\mathcal{N}(0, 1)$. Given that approximately 95% lie in the interval $(-2, 2)$, the residual plots show a random scatter of points with no clear patterns in the data, and the QQ plot and histogram are roughly aligned with what we expect under Normality (points close to a straight line in the QQ plot and an approximately bell-shaped histogram). We also see no evidence of change in variance in any of the residual plots, either for the quantitative regressors or across the levels of the qualitative regressor. Therefore, for the model we proposed, the relevant assumptions appear to be upheld.
